Functional fingerprinting stability across weeks, noise correction and atlas resolution

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Introduction

Functional fingerprinting refers to an ensemble of methods, like **Intraclass Correlation Coefficient (ICC)**, that try to identify a characteristic pattern of Functional Connectivity (FC) in subjects that share one or several features (e.g. cognitive traits, symptoms[1]).
FC fluctuations reflect **autonomous processes**, such as breathing and cardiac cycles, and some sources of noise show an idiosyncratic structure[2].
Certain effects of noise on fingerprinting are known in few sessions[3], but not in **dense sampled data**.
This study assessed the role of noise regression, atlas coarseness and number of independent sessions on functional fingerprinting.

Methods

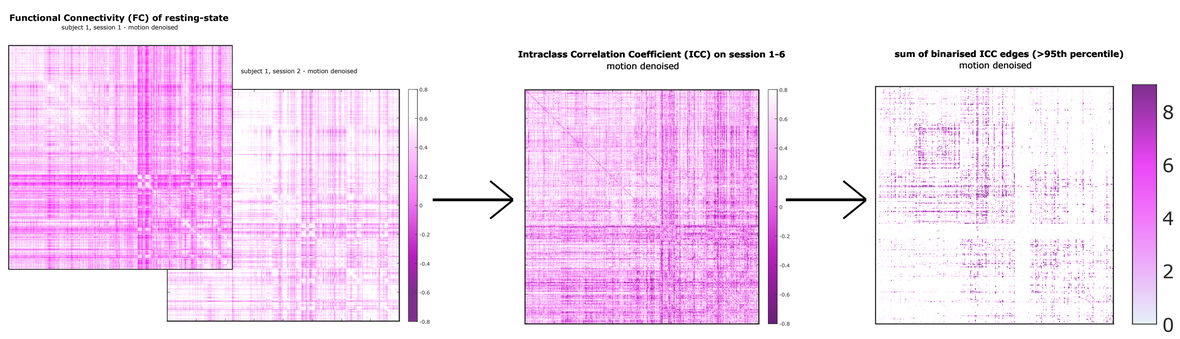
Ten healthy subjects (24-40yrs) underwent **10-minute resting-state** BOLD multi-echo sequence ten times every seven days at the same time of the day.

Anatomical and functional data were minimally preprocessed[4]. **Peak detection** for **PPG** and **chest belt** signals was run automatically, then manually corrected, with peakdet[5]. **Respiratory variance per time** (RVT), **heart rate variability** (HRV), and **16 RETROICOR regressors** were computed with phys2denoise. RVT and HRV were convolved with respiratory[6] and cardiac[7] response function.

1. motion denoised
6 realignment parameters and their derivatives + 4 Legendre Polynomials
2. motion denoised with bandpass
model 1 + bandpass 0.01 0.15
3. non-physiological tissue denoised
model 1 + CSF and WM averages orthogonalized w.r.t. HRV, RVT, and RETROICOR regressors.
4. non-physiological tissue denoised with bandpass
model 3 + bandpass 0.01 0.15
5. tissue denoised
model 1 + CSF and WM averages
6. tissue denoised with bandpass
model 6 + bandpass 0.01 0.15
7. physiologically denoised
model 5 + HRV + RVT + 16 RETROICOR regressors
8. physiologically denoised with bandpass
model 7 + bandpass 0.01 0.15

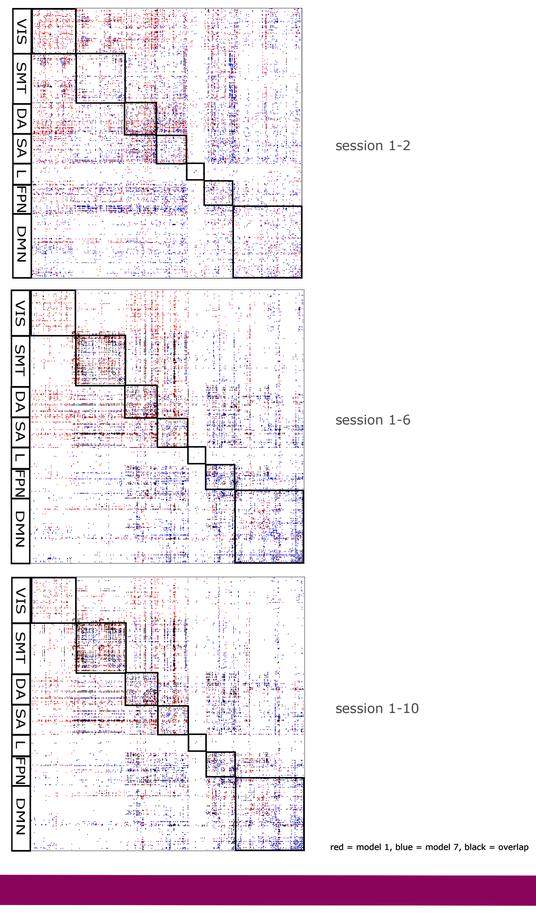
$$\frac{\mathrm{MS_R}-\mathrm{MS_E}}{\mathrm{MS_R}+(k-1)\mathrm{MS_E}+\frac{k}{n}(\mathrm{MS_C}-\mathrm{MS_E})}$$

ICC(2,1). MSR = mean square for rows (i.e. subjects); MSE = mean square for error; MSC = mean square for columns (i.e. sessions); k = number of raters/measurements (here sessions)



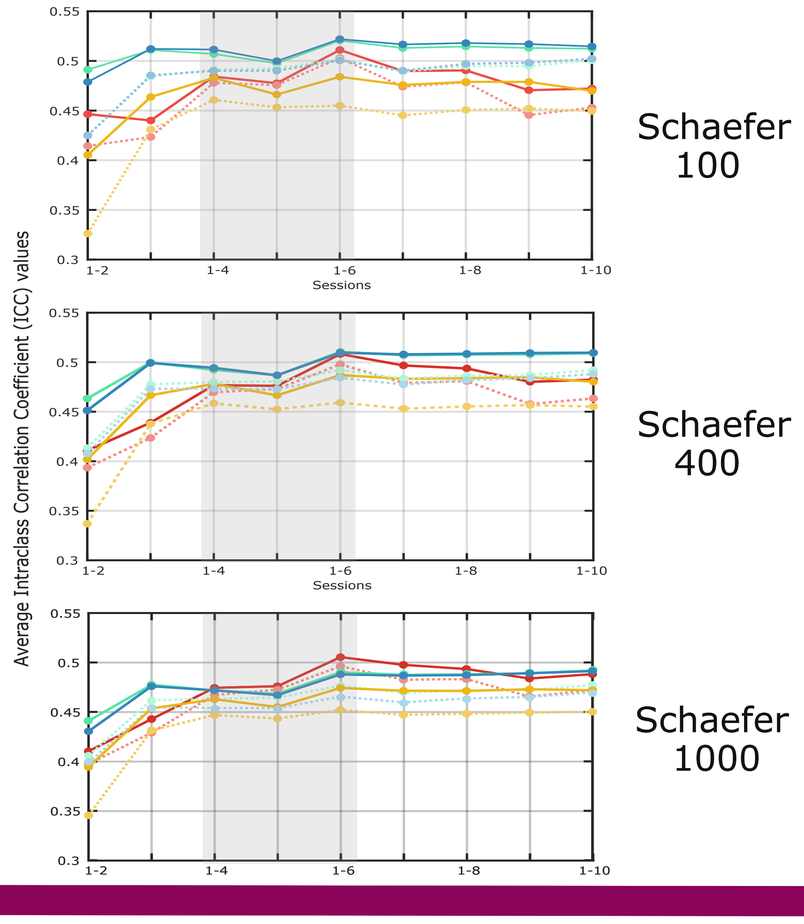
Functional fingerprinting was assessed with **ICC(2,1)**[8], as in [9], assuming lower intra-individual than group-level FC variability indicates subject distinctiveness. We computed ICC **incrementially**, with FC from **2 to 10 sessions.**

To assess the **topographical distribution of most stable edges**, we thresholded and binarised each ICC map at **95th percentile** for each denoising approach, summing all maps.

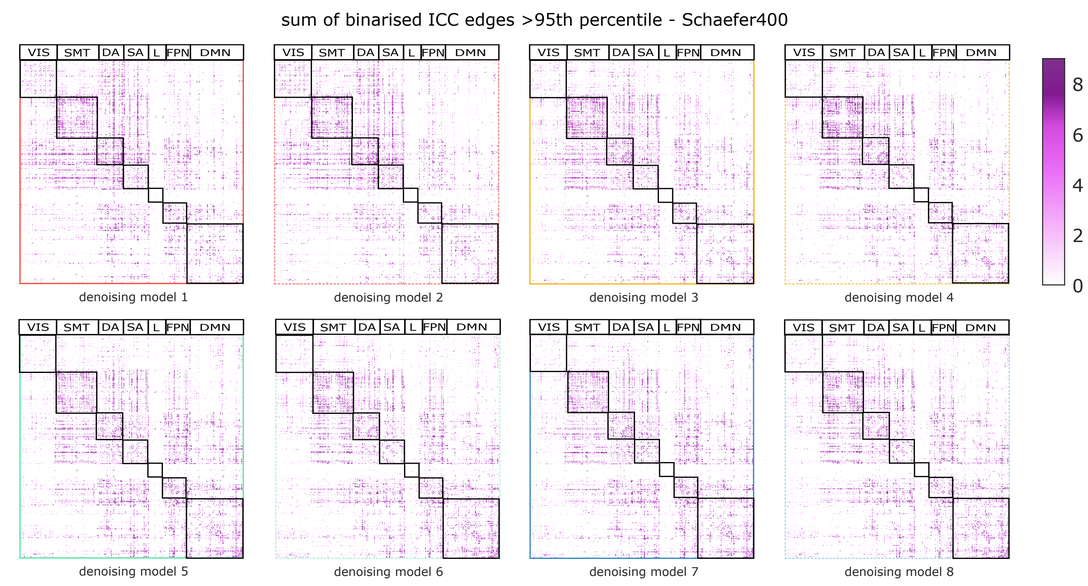


4–6 10 minutes sessions: the sweet spot for functional fingerprinting

Higher functional fingerprinting can be achieved either by acquiring more sessions - minimum 4, ideally 6 - or by applying stronger physiological noise correction, (in)dependently from atlas coarseness (Schaefer 100 vs 400 vs 1000)



Results



VIS = visual; SMT = somatomotor; DA = dorsal attention; SA = salience; L = limbic; FPN = fronto-parietal; DMN = default mode network

ICC scores were **higher** compared to previous findings[10] (mean = 0.32-0.52).

Dense-sampling definitely influences fingerprinting, with the highest ICC achieved when considering **6 sessions**, regardless of noise correction.

Tissue-denoised and physiologically-denoised data seem to provide the **most stable and reliable results** across time and atlases.

Coarser parcellations benefit more from physiological noise regression in terms of ICC, but **average ICC trends are the same** across atlases.

ICC is generally higher within **Somatomotor (SMT) and Dorsal Attention (DA) networks**, but when tissue- and physiological denoising highest ICC scores are found in the **default mode network (DMN) and fronto-parietal network (FPN)**.

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